

Bullet Farm: A Software-Change Transaction Processor for Heterogeneous Agents

Bullet Farm Maintainers

Abstract—Modern coding systems are commonly organized around an agent loop, a sandbox, a worktree, a reviewer, or a merge queue. Those mechanisms are useful but do not by themselves implement a software-change transaction. The missing contract binds who may mutate an exact repository subject, which process incarnation owns that authority, what evidence was obtained for which bytes, and how an ambiguous remote effect is reconciled without a blind retry. Bullet Farm is designed as a transaction processor around heterogeneous agents. Its binding chain is: incarnation-fenced authority, one repository writer, private Attempt clones, exact-OID Candidates, independently bound Evidence, durable effect intent, read-back and adoption after ambiguity, and truthful UNKNOWN projection across five trust planes. This paper evaluates the architecture against a dated public corpus: Gas Town v1.2.1, Gas City v1.4.1, DeepSeek Harness dsh-v0.1.1-rc.2, Omnigent v0.10.0, selected issue reports, and a broader tool landscape. It separates operational features from authority/integrity properties and uses four maturity labels: **Designed**, **Component-proved**, **Transaction-proved**, and **Live/release-proved**. At evidence snapshot 20260825-stage1, Bullet Farm has component receipts but no connected TRANSACTION_PROOF, live proof, or release proof; the release inventory reports 26 blocked of 26 evaluated gates. Accordingly, this Stage-1 paper makes no performance or universal-superiority claim. A Stage-2 revision and matched benchmark are deferred until the connected transaction proof exists. The paper and its four-page executive brief consume one versioned JSON evidence manifest and are subject to deterministic publication preflight.

Index Terms—coding agents, transaction processing, Git isolation, evidence binding, authority fencing, effect reconciliation, merge safety.

I. RESEARCH CONTRACT

Agent systems have become capable enough that their limiting failure is often not model reasoning but the authority surrounding it. Two sessions can edit one checkout; a stale reviewer can approve bytes that no longer exist; a timed-out push can be retried after the server already applied it; and a UI can turn missing state into apparent success. Classical transaction-processing work treats identity, concurrency, logging, recovery, and idempotence as one problem [1]–[3]. Coding-agent stacks frequently distribute those obligations among an editor, Git, CI, a provider process, and a forge without an end-to-end subject binding.

Bullet Farm’s thesis is deliberately narrower than “better agents.” It is a software-change transaction processor *around* heterogeneous agents. A model may reason, search, and propose, but it cannot mint its own write, verification, or integration authority [4]. The research questions are:

- RQ1: Which identities and exact subjects must be bound so that a concurrent agent run can be recovered without inferring completion?
- RQ2: Which bindings are implemented and component-proved in the current family, and which remain design or connected-product gaps?
- RQ3: Do the reviewed pinned public contracts of adjacent systems document the same end-to-end bindings, after separating operational capability from authority and integrity?
- RQ4: What observations would falsify the design, and what matched experiment would be required before a comparative performance claim?

A. Inclusion and Source Discipline

Deep studies were included when a system exposed a public architecture plus a release or commit suitable for reproducible inspection. The fixed subjects are Gas Town v1.2.1, Gas City v1.4.1, DeepSeek Harness dsh-v0.1.1-rc.2, and Omnigent v0.10.0 [5]–[8]. Issue reports were included only when they described a concrete authority, Git, evidence, or recovery failure and could be mapped to a named invariant. The wider landscape appendix uses release/commit pins for open-source systems and official-document observation dates for proprietary ones. These are public-contract observations, not reverse engineering. Codex statements rely only on official OpenAI documentation [9], [10].

Every comparison cell uses exactly one of five terms. Documented means a pinned primary source describes the property. Partial/configuration-dependent means the property depends on configuration or covers only part of the row. Not documented means the reviewed sources do not describe it; it does not assert that the implementation lacks it. Unknown means the public material was insufficient to classify. N/A means the property is outside the product’s stated role. Absence of public documentation is never converted into a categorical “No.”

B. Evidence Vocabulary and Claim Limit

Table I is the paper’s claim lattice. The terms intentionally avoid the ambiguous word “complete.” They are generated into both documents from `evidence.json`, whose schema is `bullet.paper-evidence.v1` [11].

The snapshot records BLOCKED doctor status, PASS for the fast lane, PASS for the demo, and PASS for 2 bounded formal models. The release result is written as “26 blocked of 26 evaluated gates,” never as a bare fraction that could be mistaken for success. UNKNOWN, timeout, skipped, unsupported,

TABLE I
MATURITY LABELS USED THROUGHOUT THIS PAPER.

Label	Admitted meaning
Designed	Reviewed contract or invariant; no executable receipt required.
Component-proved	Exact committed component and mapped local receipt; no cross-process transaction inference.
Transaction-proved	One exact connected offline five-plane transaction with independent subject-bound receipts.
Live/release-proved	Admitted live provider/forged transaction, or signed release/install/recovery proof from the same subjects.

zero tests, a model assertion, HTTP 2xx, or process exit zero is not verified engineering truth [12]. No result from an active dirty tree is admitted into the committed evidence manifest.

II. THREAT AND FAILURE MODEL

The V1 threat model is local-first and single-user. It treats model output, repository content, concurrent agent incarnations, retrying supervisors, provider subprocesses, and network responses as untrusted. It trusts the host’s admitted binaries and kernel primitives within explicitly tested boundaries; remote multi-tenant runners and Byzantine forge operators are out of scope. The same local UID remains a material limitation because it can race path-based tools that have not been sealed or made descriptor-relative.

The recurring failures are not peculiar to agents. They are ordinary distributed-systems failures expressed through Git:

- 1) *Authority aliasing*. Two living processes believe they own the same Attempt or writable checkout. A session name is not an incarnation fence.
- 2) *Repository aliasing*. Linked worktrees have per-worktree HEAD and index, but share the object database, common metadata, and most refs [13]. They improve workspace ergonomics; they do not create separate repository principals.
- 3) *Subject drift*. Tests or reviews name a branch, workspace, or PR rather than the exact base, head, tree, patch, environment, and gate set. Rebase or merge can silently invalidate the result.
- 4) *Ambiguous effects*. A timeout occurs after a server may have accepted a push or workflow creation. Retrying before read-back can duplicate the logical mutation.
- 5) *Inferred liveness*. tmux, a PID, an idle terminal, or a model’s “done” statement becomes a completion oracle. Silence is then confused with death or success.
- 6) *Destructive recovery*. Cleanup removes a workspace before durable preservation and read-back establish whether the last mutation happened.
- 7) *Projection promotion*. A dashboard converts absent, stale, or contradictory ledger state into a green label. The UI becomes accidental authority.

The public issue corpus supplies concrete instances: town-root checkout deleted operational files [14]; parallel rig-root writers clobbered edits [15]; a timeout after a server commit admitted duplicate workflows [16]; and a non-atomic claim let two sessions execute the same task [17]. These reports are evidence for failure classes, not proof that the projects remain vulnerable at later heads.

A. The Transaction Subject

The minimum authority subject is not “agent a edits repository r .” A useful abstraction is

$$A = (m, g, v, a, i, f, w, p, b),$$

where m is an immutable Mission revision, g its graph revision, v a Variant, a an Attempt, i the process incarnation, f its monotonically ordered fence, w a workspace generation, p a policy generation, and b a bounded budget reservation. A mutating request that omits one of these fields can accidentally alias a different logical run. An expired lease does not merely change a timestamp: reclaim changes i and f , so every old request is stale even if its bearer still possesses a valid-looking process handle.

The engineering subject is similarly richer than a commit hash:

$$C = H(\text{domain}, A, o_{\text{base}}, o_{\text{head}}, o_{\text{tree}}, h_{\text{patch}}, e, p),$$

where the OIDs are algorithm-tagged, e identifies admitted environment and tool subjects, and all variable-length inputs are framed. The formula is a design description, not a claim that every current constructor already carries every term. It explains the invariants: equal content with different ancestry or authority need not be the same Candidate; a moving branch is not a Candidate; and a verification result for C says nothing about C' after a rebase, merge, policy change, or gate change.

An effect has its own identity rather than borrowing the Candidate identity:

$$X = (C, E, r, o_{\text{expected}}, o_{\text{desired}}, k, f),$$

where E is the independent Evidence root, r is the protected destination, k is the logical idempotency key, and f is rechecked at the broker. The desired Git OID can be observed independently; the transport response cannot. The ledger therefore distinguishes command acceptance, local application, verification, effect intent, remote observation, and projection. Collapsing those into one “done” bit destroys the information needed for recovery.

B. Trust-Boundary Attacks

The model is intentionally hostile to ambient convenience. Repository files may request clean/smudge filters, external diff programs, merge drivers, credential helpers, hooks, submodules, alternate object stores, filesystem monitors, pagers, editors, or URL rewrites. Provider output may contain terminal escape sequences, unbounded JSON, paths outside scope, or shell names disguised as gate requests. A runner may deliver a valid request twice, reorder a stale heartbeat after reclaim, or crash between local mutation and response. A forge may apply a write and lose the response. A projection may reconnect behind its watermark. Each is treated as a typed protocol case, not an instruction to “try again.”

Conversely, Bullet Farm does not claim to defeat an administrator who can replace the running kernel, signing roots, or admitted binaries. It does not make a same-UID path race impossible where an external tool still resolves a mutable path. The design shrinks and names the trusted computing

TABLE II
FAILURE CLASSES, UNSAFE SHORTCUTS, BINDING RULES, AND EXTERNALLY VISIBLE REFUSAL.

Failure class	Unsafe shortcut	Required binding	Honest observable outcome
Authority aliasing	PID, tmux name, or lease expiry treated as ownership	Attempt + incarnation + fence checked at every mutation	Stale principal is refused; current work remains unaffected
Repository aliasing	Parallel sessions write one checkout or common refs	Sole daemon writer and private generation per Attempt	Second writer has no repository capability
Subject drift	CI/reviewer names branch, PR, or working directory	Candidate exact base/head/tree/patch/environment; Evidence names Candidate	Rebase/merge/gate change invalidates Evidence
Ambiguous effect	Timeout classified as failure and retried as new work	Durable effect intent, one idempotency key, expected-old-OID CAS, read-back	UNKNOWN until exact applied/absent/divergent observation
False liveness	Idle terminal or missing heartbeat classified as completion/death	Lease/fence plus reconciled process and ledger observations	ALIVE, DEAD, UNKNOWN, or CONTRADICTORY
Destructive recovery	Force-remove before preserving unique bytes	Sealed preservation subject and durable cleanup/tombstone outcome	Cleanup refuses or returns UNKNOWN; recoverable bytes remain
Projection promotion	UI infers green from missing/stale state	Sequence/watermark-bound projection with no authority methods	Stale/absent/contradictory state remains visible

base; it does not abolish one. Release acceptance therefore includes binary-digest admission, descriptor-relative/no-follow mutation, signed packages, provenance, and platform-specific containment rather than assuming that a source build is an installer.

III. FIVE-PLANE TRANSACTION ARCHITECTURE

Figure 1 shows the complete binding chain. The five planes are separated so that no plane can use its own assertion as independent evidence.

A. Control and Execution

The control plane owns Missions, immutable graph revisions, commands, leases, fences, policy snapshots, and durable outbox intent. A lease is insufficient without a monotonically changing incarnation fence checked by every mutation gateway [18], [19]. Reclaiming expired work produces a new authority subject; the old process cannot regain authority by waking up.

The execution plane supervises heterogeneous providers through admitted, bounded protocols. A provider receives read context and returns a typed patch proposal. It does not receive repository authority. Exactly one BulletGit daemon is the repository writer. Each Attempt uses a private clone rather than a linked worktree, eliminating shared index/common-ref control paths between Attempt principals. Hostile repository configuration, hooks, alternates, credential helpers, and environment are treated as input to be cleared or refused, not inherited ambient authority [20].

B. Candidate and Verification

Applying a validated proposal produces an immutable Candidate identity over the provenance needed to reproduce it: base OID, head OID, tree OID, canonical patch identity, Attempt/fence, and relevant environment and policy subjects. A content hash alone is insufficient because identical bytes can arise under different authority or ancestry. A branch name is insufficient because it can move.

The verification plane reconstructs that exact Candidate in an independently clean environment. Gates come from an admitted catalog, not model-provided shell. Evidence binds

the Candidate and gate IDs, executable/environment subjects, bounded output, exit status, and timing. Any subject change after verification invalidates the Evidence. A reviewer may be a different model or a deterministic tool; independence is an authority relation, not a persona label.

C. Delivery, Audit, and Ambiguity

The delivery plane alone holds forge credentials. Before an effect it durably records logical effect identity, expected-old OID, intended-new OID, Evidence root, and idempotency key. Protected integration tests the exact merge result, following the merge-queue principle that what is tested should be what becomes the branch [21], [22]. GitHub publication, when enabled, is a one-way mirror rather than a second writer: expected-old-OID CAS, read-back, divergence quarantine, and independent backups.

If transport becomes ambiguous, the command remains UNKNOWN. The broker reads the authoritative ref or remote object. An exact match can be adopted with a receipt; a mismatch is quarantined; absence may admit a retry under the same logical effect identity. HTTP success is not verification, and transport failure is not proof of non-application. This is the idempotent receiver discipline applied to Git effects [3].

The audit plane projects append-only events and exact watermarks. Portal state may be PENDING, APPLIED, VERIFIED, FAILED, or UNKNOWN; stale or contradictory streams remain visibly stale or unknown. It cannot mint authority, repair evidence, or reinterpret missing data as success.

D. Worked Failure and Recovery Transaction

Consider the failure class in Gas City issue 5511, where a timeout after a server-side commit led to additional workflows [16]. In the Bullet design: (1) a command and one effect identity are committed before dispatch; (2) a fenced Attempt in a private clone yields Candidate C ; (3) an independent verifier yields Evidence $E(C)$; (4) the broker records an intent to advance protected ref r from o to n ; (5) the network times out; and (6) the command becomes UNKNOWN, not failed and not complete. Read-back then has three outcomes. If $r = n$, the broker adopts the applied effect. If $r = o$, it may retry the same effect identity. If r is neither, integration is quarantined as divergence. A new runner incarnation cannot create a second

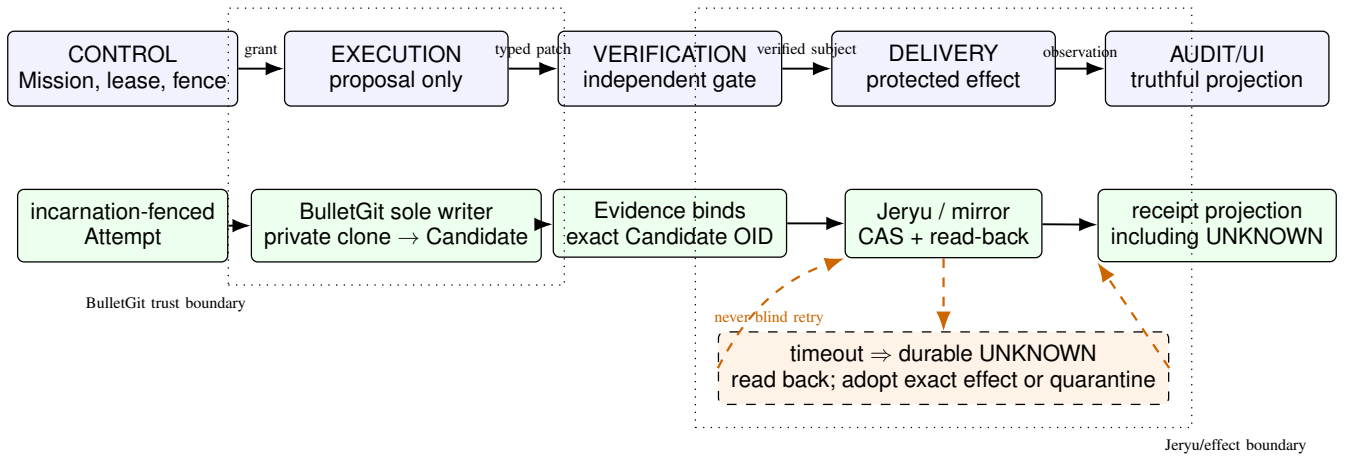


Fig. 1. Five trust planes and the cross-plane binding chain. A timeout after effect intent produces durable UNKNOWN; read-back either adopts the exact observed effect or quarantines divergence. It never authorizes a blind retry.

logical effect because the ledger, fence, expected-old OID, and idempotency key are all checked at the mutation boundary.

The chain is the product: incarnation-fenced authority → sole repository writer → private Attempt clone → exact-OID Candidate → independently bound Evidence → durable effect intent → read-back/adoption → truthful projection. A sandbox, reviewer, or merge queue can strengthen one link but does not imply the chain.

E. Cross-Plane Invariants

The architecture is constrained by invariants rather than by agent roles. The current registry distinguishes schema controls, gateway controls, and tests so that prose cannot be the only location of a safety rule. The following ten are the compact paper view; the generated registry and closure plan remain the authoritative crosswalk [19], [23].

- 1) **One current incarnation.** At most one live incarnation/fence may hold mutation authority for an Attempt. Lease expiry alone does not grant the successor authority; the successor's durable fence does.
- 2) **One repository writer.** Provider, runner, verifier, portal, and effect processes lack direct workspace mutation capability. Only BulletGit accepts a validated operation under the exact reservation.
- 3) **Private repository generations.** Attempt generations do not use linked-worktree `.git` files or share mutable repository control state. Prior-or-complete-next publication is preserved across crashes.
- 4) **Scoped atomic patch.** Every operation is validated before any write. Traversal, `.git`, symlink, case-fold, normalization, duplicate, special-file, body-size, and operation-count ambiguity refuse the entire batch.
- 5) **Candidate identity is immutable.** Base, head, tree, patch, and authority/provenance subjects are length-framed and domain-separated. Content identity and provenance-bound Candidate identity are not conflated.
- 6) **Evidence is non-transferable.** The verifier reconstructs an exact Candidate with admitted gates. Any change to

Candidate, gate, environment, policy, or verifier subject makes prior Evidence inapplicable.

- 7) **Attensor is not broker.** The principal that produces Evidence cannot use that act to mint protected-ref authority. The effect broker checks Evidence and current control authority independently.
- 8) **Effect identity survives retry.** Intent is durable before transport. Expected-old OID, desired OID, idempotency key, and fence are read-back/adoption subjects, so retry never creates a second logical effect.
- 9) **Preservation precedes destruction.** Unique workspace bytes, journal identity, destination, and cleanup target are sealed and rechecked. Post-delete ambiguity remains an outcome, not retrospective success.
- 10) **Projection is non-authoritative.** Commands return pending before settlement. Sequence gaps, conflicting events, absent routes, and unavailable observations yield stale or unknown state; UI actions still pass through the authenticated command boundary.

These invariants deliberately retain separation among three sovereign truths. The Kernel decides whether an operation is authorized. BulletGit decides what repository subject exists. The verifier decides whether Evidence for that subject satisfies admitted gates. The broker decides whether an authorized, verified subject may be proposed to the forge, while the forge remains sovereign over its protected ref. No single database row replaces those independent observations.

F. BulletGit as the Repository Trust Boundary

BulletGit is not intended to replace Git's object model or pack protocol. It wraps ordinary Git with a capability and evidence boundary. A bare mirror can amortize network/object transfer, but an Attempt clone dissociates before use so no surviving alternates file makes mirror garbage collection part of workspace correctness. The clone loses its origin and ambient authentication. Git child processes receive a scrubbed environment and explicit configuration that disables inherited global/system config, paging, hooks, credentials, external attributes, signing, editors, and prompts.

Repository-scoped commands first inspect local configuration with includes disabled; command-bearing or truth-redirecting values fail closed [20].

This distinction matters because a repository is active input. A clean filter can execute during checkout; `textconv` can alter what a diff appears to show; an external merge driver can create bytes outside the intended patch; URL rewriting or a remote helper can redirect network authority; a credential helper can make a private clone push-capable; alternates can make object availability depend on another repository; and hooks can turn an apparently read-only Git verb into execution. A provider sandbox does not automatically neutralize those semantics, especially when the provider itself runs Git.

Patch application validates the complete batch before mutation. Paths are canonical normalized relative bytes with no parent traversal, `.git` component, NUL, backslash ambiguity, alternate-data-stream syntax, trailing dot/space, symlink traversal, or folded collision. Operation and body counts are bounded. Checkpoints use a temporary index rather than the live workspace index. A Candidate is minted only after Git reports the expected tree/head relationship and the journal durably identifies the generation.

Preservation is part of the protocol rather than cleanup hygiene. A sealed receipt binds source workspace, recoverable artifact, destination, journal, Candidate/Attempt subjects, and exact cleanup target. Immediately before deletion the daemon re-verifies the receipt and the filesystem object. A synced tombstone records the settled cleanup outcome. If failure occurs after deletion but before confirmation, the result is typed UNKNOWN; the implementation does not reconstruct certainty from absence.

The current production constructor is intentionally fail-closed: positive online final-authority settlement is unavailable, so the public mutation path refuses before repository I/O. That refusal is useful component evidence. It is not evidence that the designed positive path works. Enabling it requires an immutable signed shared-wire subject, locally verified grants, current Kernel reservation/fence read-back, short operation permits, and durable settlement of the same transaction.

G. Jeryu Source Authority and Blocked Topology Proposals

The frozen V1 topology distributes source Jeryu-only from immutable signed tags. It assumes neither `git.neverhuman.org` nor any GitHub namespace or public mirror, and the family manifest admits no authenticated Jeryu URL. The only permitted local Jeryu family is `/home/ubuntu/jain-split/jeryu-split`; `/home/ubuntu/jeryu-split` must never be recreated.

The named `git.neverhuman.org` deployment and `github.com/neverhuman/jeryu` and `github.com/neverhuman/jankurai` front doors are unratiﬁed, operator-blocked proposals, not current topology or authority. If an operator later ratiﬁes a public mirror, GitHub remains a separately configured effect adapter rather than source authority. Mirroring must then be one-way and receipt-bearing: observe the remote ref, require the expected-old OID, publish the exact Jeryu object, read

it back, and quarantine divergence. Backups are proved separately because a mirror is not a backup.

Jeryu’s local hub provides branded onboarding, the family manifest, and pinned component mapping while component repositories remain independent. Jankurai is different: it is an existing monorepo, but no public front door is assumed. Before any external-history or mirror decision, the transaction rule is preservation first—complete ref and object bundles, ancestry map, reconciliation RFC, rehearsal, and explicit approval. A force-push would discard the very evidence required to decide which history should survive. The local-family rule is frozen. Endpoint and mirror proposals remain Designed and operator-blocked unless dated ratiﬁcation precedes TLS, authentication, protection, push/check/read-back, backup/restore, deployment, and mirror receipts.

IV. IMPLEMENTED EVIDENCE AND CONNECTED GAPS

Table III separates current component receipts from the connected claims they do not establish. Repository subjects are drawn from the shared evidence snapshot: Hub `f33750c14112`, Kernel `419d8a6832ee`, BulletGit `236f4efe2b82`, and Portal `95108e37d643`. A later source commit invalidates mapped receipts until replay; publication-only descendants are admitted only when preflight proves that evidence-critical inputs did not change.

The table is intentionally asymmetric. Numerous component tests can coexist with zero connected-product proof. `just demo` is a deterministic demonstration, not a live transaction. The doctor result is BLOCKED because the checked family lock is intentionally unsupported and stale. The release inventory is a truthful negative receipt, not a score.

A. Evidence Lifecycle

An admissible receipt answers five questions: *who observed what exact subject, using which admitted mechanism, under which boundary, with what outcome?* Test counts alone answer none of them. The paper snapshot consequently stores repository commit and tree, clean-state fact, tool versions, complete command, exit code, semantic outcome, and a sentence stating what the result does not prove. It also hashes the family manifest, family lock, release truth, invariant registry, formal locks, policy, and competitor snapshot. These inputs are deliberately outside the PDF hash loop.

Evidence moves monotonically only within one immutable subject. A new commit, tree, policy generation, executable, gate, environment, or dependency lock does not make the old receipt false; it makes it inapplicable to the new subject. Replaying a command in a dirty tree is useful diagnosis, but its result cannot be committed as frozen evidence because the tested bytes are not named by the repository commit. Similarly, a generated release report is a projection of registered receipts and blockers. It cannot satisfy one of its own rows.

Component receipts are still valuable. Atomic Kernel tests can falsify a lease or command implementation. Hostile Git fixtures can falsify configuration or path handling. Browser tests can falsify projection behavior. Bounded model checks can falsify protocol invariants under their state bounds. The

TABLE III
IMPLEMENTED NOW VERSUS NOT YET PROVED AT SNAPSHOT 20260825-STAGE1.

Boundary	Maturity	Implemented evidence	Not established
Wire and admission	Component-proved	Canonical signed DTOs, bounded IPC, executable-digest and launch-grant components	Immutable shared-wire publication and connected issuer/consumer settlement
Control authority	Component-proved	DB-clock leases, fences, atomic command/outbox, authenticated loopback ingress, truthful reconciliation	One signed authority path across Kernel, Runner, and BulletGit
Provider boundary	Component-proved	Four offline message subsets, policy-first refusal, bounded process and Linux egress components	Any live provider receipt; policy remains live-disabled in the frozen evidence subject
BulletGit	Component-proved	Fail-closed authority, private-clone/preservation primitives, reservations, cleanup UNKNOWN, strict local manifests	Positive online sole-writer transaction and production Jeryu authority
Verifier	Component-proved	One admitted fixture gate, exact-subject Evidence, bounded transport failure handling	Production reconstruction source, executable digest for every gate, and multi-gate aggregation
Effects	Component-proved	Command/effect ambiguity model and local bare-forge components	Authenticated Jeryu/GitHub expected-old-OID push and read-back receipts
Portal/audit	Component-proved	Generated contracts, authenticated command reconciliation, watermark-bound projections, explicit absent-surface UNKNOWN	Packaged runtime and complete Candidate/Evidence/Effect ledger subjects
Security audit	Component-proved	Deterministic score 58 (raw 58), 10 caps, 29 hard and 15 soft findings	V1 floor ≥ 90 , zero caps/hard findings, and a portable pinned hosted artifact
Connected product	Designed	Normative five-plane transaction and release contracts	Transaction-proved or Live/release-proved evidence; 26 blocked of 26 evaluated release gates

TABLE IV
STAGE-1 COMMAND FAMILIES, RECORDED OUTCOMES, AND INTERPRETATION. FULL COMMANDS AND BOUNDARIES ARE IN `EVIDENCE.JSON`.

Command family	Snapshot outcome	What it establishes	What it cannot establish
bullet-family doctor -json	BLOCKED	Fail-closed install/lock diagnosis	A supported installed family
bullet-family check release -json	BLOCKED	26 named blockers among 26 evaluated gates	Any positive gate or percentage score
Hub just fast	PASS	Mapped Hub component checks at the subject	Sibling, transaction, live, or release truth
Hub just demo	PASS	Deterministic multi-process demonstration semantics	TRANSACTION_PROOF; provider/forge/release evidence
Hub contract lane	PASS	Wire/schema generation and drift checks	Immutable publication or consumer settlement
Invariant registry	PASS	Machine crosswalk and enforced-entry validation	Implementation of rows whose lifecycle is planned
Local audit lane	PASS component; release blocked	Score 58/raw 58, 10 caps, 29 hard and 15 soft findings	V1 quality floor or portable hosted execution
Repository required lanes	PASS	Mapped committed components for Kernel, BulletGit, and Portal	Cross-repository composition
Formal model lane	PASS	2 bounded models match pinned sources/tools/counts	Liveness or implementation correctness
Paper build/preflight	deterministic publication check	Shared values, citations, layout, metadata, embedded fonts, page size/count, reproducible PDF bytes	Product correctness or release authority

discipline is to preserve those results without performing an invalid conjunction: “Kernel passed” $\wedge$ “BulletGit passed” $\wedge$ “Portal passed” does not imply that a signed command traveled through all of them with one Candidate and one observed effect.

B. Formal Models and Their Boundary

Exactly two bounded TLA+ models are pinned: lease/fence/reclaim and command/effect ambiguity under timeout [24]. At the snapshot the model lane is PASS for 2 models at the recorded tool, source, configuration, and state counts. These are safety-model component receipts. They do not model Git implementation, provider protocols, the forge, OS containment, liveness/fairness, multi-tenant attackers, or the entire product. Deadlock checking is not promoted into a claim that the service always makes progress. Model success therefore remains Component-proved, never Transaction-proved.

C. Open Connected Product Boundary

The authoritative closure plan, not this paper, controls V1 [19]. The remaining predecessor chain includes immutable

wire publication, signed online authority, connected Candidate/Evidence settlement, authenticated protected integration, provider conformance, packages, installer, backup, signing, provenance, and live/release receipts. The new workplan is only an opportunity backlog and explicitly loses to that authority [25]. Most importantly, the absence of an end-to-end TRANSACTION_PROOF is not hidden behind component totals.

V. PINNED SYSTEMS AND CAPABILITY MATRICES

The comparison asks two different questions. Operational capabilities concern what work a system helps users perform. Authority/integrity properties concern what exact subject and principal the public contract binds. Combining them would incorrectly make a rich orchestrator look unsafe merely because it is featureful, or make a strict component look operationally complete merely because it refuses unsafe work.

A. Deep Case Studies

Gas Town v1.2.1. Gas Town provides a legible multi-agent town: role-specific agents, Beads-backed work, tmux sessions,

hook worktrees, patrol and recovery machinery, and a Refinery merge queue [5], [26]. Its strength is operations: many agents and providers can be supervised using durable work records. Its public contract uses worktrees and orchestration roles rather than a separate repository-writer principal. Because linked worktrees have their own HEAD and index but share the common Git repository and most refs, they are not equivalent to private repository principals [13].

Gas City v1.4.1. Gas City generalizes the factory with Agents, Beads, Formulas, Rigs, Packs, Events, multiple runtimes, retry/fan-out/drain controls, and release artifacts [6], [27]. The richer runtime and workflow vocabulary should not be minimized. The issue audit nevertheless shows why session durability and exactly-once effects are distinct: a shared rig root, a non-atomic task claim, and retry after ambiguous creation all cross different trust boundaries [15]–[17].

DeepSeek Harness rc.2. DeepSeek Harness is a Cordis plugin kernel: model, tools, sandbox, storage, session, loop, scheduling, and UI are composed as plugins, with model-visible context reconstructed from an append-only session log [28]. It is not accurately described as categorically single-agent. Its experimental Agent Teams seam adds a roster, task board, mailbox, and continuable child agents. The same-process team documentation also warns that checkout scopes are advisory rather than locks [29]. Its primary contribution is composable agent execution, not protected Git integration.

Omnigent v0.10.0. Omnigent wraps several vendor harnesses behind a uniform session/runtime layer. It documents Linux bubblewrap and macOS seatbelt sandboxes plus an L7 proxy that keeps credentials outside the sandbox and can transform network requests [30], [31]. Contextual policy verdicts are ALLOW, DENY, and ASK; transformation belongs to the L7 proxy, not the policy verdict [32]. Omnigent Polly adds parallel worktrees and cross-vendor implementation/review, which is a meaningful operational feature [33], [34]. The wrapped harness still performs repository actions, so approval attribution and Candidate binding remain separate questions.

B. What the Architecture Should Borrow

The comparison is not a feature-stripping exercise. Gas Town demonstrates the operator value of named roles, inspectable work records, patrol, recovery, and a dedicated landing function. Gas City makes runtimes and workflow structures more explicit and adds release-operational depth. A transaction processor that cannot explain where work is, recover a session, limit dispatch, or expose a usable queue will be safer only by being unusable. Bullet’s control plane should compose these operational concepts while ensuring that a role name never substitutes for an authority subject.

DeepSeek Harness shows that provider, tools, storage, sandbox, loop, and UI can be composable plugins without turning every integration into a special case. Its append-only model-visible session log is especially relevant: if context cannot be reconstructed from durable events, recovery silently changes the agent’s inputs. Bullet can consume a plugin kernel behind the provider boundary as long as plugin composition cannot acquire repository or effect authority. Experimental Teams also

demonstrates that subagent coordination and a task board can be first-class protocol objects. The remaining difference is enforcement: advisory checkout scopes are useful coordination, while a transaction gateway must refuse overlapping mutation.

Omnigent demonstrates mature defense in depth at a different layer. Native OS sandboxing, a credential-isolating L7 proxy, contextual approval, shared sessions, and cross-vendor review are desirable even when sole-writer and Candidate bindings exist. Bullet should not attempt to reimplement every agent UX. It can wrap or accept proposals from those systems, provided the proposal is bounded untrusted data and all repository/effect credentials remain across the gateway. Conversely, an L7 proxy can transform or authenticate network traffic without becoming the authority that says a Candidate passed its gates.

These examples suggest a clean interoperability rule: protocols such as ACP and MCP may carry sessions, context, and tool calls, but they do not mint Bullet capabilities. A provider adapter may translate a typed proposal. A review adapter may return commentary. An MCP tool may expose read-only repository facts. None may cross from “the model requested” to “the repository is authorized” without the same signed control subject, current fence, admitted operation, and durable settlement used by native components.

C. Classification Procedure

For each cell, the review first fixed the release/commit and source set; then looked for an explicit contract; then assigned the least affirmative supported term. A feature shown only in an optional pack, plugin, hosted configuration, or experimental seam is **Partial/configuration-dependent** unless the row itself is about that seam. **Not documented** records a search result, not an implementation claim. **Unknown** is used when the public description is too ambiguous even to say that the property is undocumented. **N/A** is reserved for products whose public role does not include the property—for example, a plugin kernel that does not claim to operate a protected forge.

The Bullet design and Bullet Stage-1 rows are separate. Otherwise a design document would be compared against released competitors as if it were running code. The current row is **Partial/configuration-dependent** whenever some exact component exists but the positive connected path does not. A fail-closed unavailable production constructor is evidence that unauthorized writes are refused; it is not evidence of an operational repository writer. This conservative classification also prevents the matrices from becoming marketing scores: the terms are not ordered numerical values and should not be summed.

D. Why Familiar Mechanisms Do Not Automatically Compose

One objection is that every part of the chain already exists: worktrees isolate files, containers isolate processes, CI verifies commits, merge queues protect branches, and idempotency keys prevent duplicate requests. The premises are mostly correct; the conclusion does not follow unless the subjects and principals at their boundaries are the same.

A worktree is not a writer identity. It gives a convenient per-worktree HEAD and index, yet common repository metadata and most refs remain shared [13]. An agent that can invoke arbitrary Git can still name refs, helpers, remotes, hooks, config, objects, and sibling worktrees outside the intended edit surface. A private clone with authentication removed is not magical isolation either, but it makes the repository principal a separable object that the sole writer can validate and destroy.

A sandbox is not a repository transaction. A container, bubblewrap, seatbelt profile, or remote VM can restrict filesystem and network access. It does not decide whether the bytes being written are an admitted patch for the current fence, whether tests name the resulting tree, or whether a protected push applied after timeout. Sandboxing is a necessary defense against provider and repository content; it complements rather than replaces Candidate and effect identity.

A durable session is not durable engineering state. Transcripts, mailboxes, task boards, tmux sessions, and event logs make reasoning resumable. They may prove what an agent saw or said. They do not prove that the repository contains a particular tree, that a gate ran against it, or that the forge accepted it. Bullet should retain session history for cognition while deriving repository truth from Git and verification truth from Evidence.

A reviewer is not automatically an independent attestor. A second model can catch semantic mistakes and cross-vendor review can reduce correlated failure. Independence still fails if both reviewers share a mutable checkout, if the author can rewrite the test oracle, if the review result names only a branch, or if the reviewer credential can execute the protected effect. The authority requirement is reconstructible exact-subject verification and attestor/broker separation; model diversity is an additional quality control.

A merge queue is not effect reconciliation. Testing the exact merge result is a central integrity property [21], [22]. The queue still needs a caller identity, current authority, an Evidence subject, expected-old ref, and a rule for lost responses. A queue item reported as failed after its merge applied can be duplicated by an upstream retry unless logical effect identity survives transport ambiguity.

An idempotency key is not useful unless the receiver binds it. A random client token stored only in process memory disappears on restart. A receiver that records the token after rather than atomically with its effect leaves a crash window. A token reused for a different desired OID launders a conflict. The effect record must bind the logical command, Candidate, Evidence, protected ref, expected and desired OIDs, current fence, and observed outcome, durably before and after transport [3].

Bullet Farm’s novelty claim is therefore compositional and falsifiable, not that it invented locks, hashes, queues, or idempotence. If a smaller public contract can demonstrate the same bindings under the fault corpus, Bullet’s extra machinery is unnecessary. Until then, comparing one isolated mechanism with the entire chain is a category error.

The result is not that Bullet Farm is universally superior. It has less live product evidence than every mature system in this table. The defensible result is that the reviewed pinned public

contracts do not document the same end-to-end bindings. Whether those bindings justify their cost is precisely the Stage-2 benchmark question.

VI. ISSUE-TO-INVARIANT AUDIT

Table VII records state and affected/reported versions as observed on 2026-08-25. “Pinned corroboration” distinguishes issue testimony from code/release material at the selected subject. An issue being closed does not erase the failure class; an open issue does not prove current pinned code is affected.

The audit corrects four tempting misreadings. First, issue 742 is about a settings path; it supports no respawn-count claim. Second, DeepSeek Teams makes categorical “single-agent” language false. Third, Omnigent policy returns ALLOW/DENY/ASK; proxy transformations are a different layer. Fourth, linked worktrees do not share one index: each linked worktree has its own index and HEAD, while common repository metadata and most refs are shared [13].

VII. EVALUATION, FALSIFICATION, AND ARTIFACT

A. Stage-1 Evaluation Protocol

Stage 1 is architecture and component assurance. From one dated evidence snapshot, the artifact records repository commit/tree/clean facts, toolchain versions, full commands, doctor and release-gate outcomes, component and formal receipts, external pins, and hashes of evidence-critical inputs. Both PDFs consume generated macros from that JSON. The strict preflight verifies the schema, subjects, trees, cleanliness, input hashes, citations, references, layout warnings, metadata, embedded fonts, letter page size, page counts, and byte-identical rebuilds at one source epoch. It deliberately fails while any canonical family checkout is dirty.

The reproducible command surface from the Hub is:

```
just paper just paper-check
cargo run -locked -quiet -bin
bullet-family - doctor -json
cargo run -locked -quiet -bin
bullet-family - check release -json
just fast just demo
```

Repository-mapped required/contract/formal commands and their complete observed outcomes are enumerated in evidence.json; summary macros do not replace those command records. The Hub evidence subject excludes publication-only descendants to avoid an impossible self-hash loop. Preflight permits that narrow descendant only when every changed path is in the paper/tooling set and all hashed evidence inputs remain identical.

B. Fault-Oriented Acceptance Corpus

A happy-path patch is a weak discriminator: nearly every reviewed system can edit a file and run a test. The connected Bullet acceptance corpus must inject faults at every state boundary and retain the resulting ledger, Git, Evidence, effect, and observation subjects. At minimum it includes:

- 1) reclaim an Attempt while its old provider and runner remain alive, then deliver old mutation, heartbeat, verification, and effect messages;

TABLE V
OPERATIONAL CAPABILITIES IN PINNED PUBLIC CONTRACTS. THESE CELLS DO NOT IMPLY AUTHORITY SAFETY.

System	Orchestration	Provider composition	Isolation	Durable sessions	Review	Merge queues
Bullet Farm Stage 1 [4], [19]	Documented	Partial/configuration-dependent	Partial/configuration-dependent	Documented	Partial/configuration-dependent	Partial/configuration-dependent
Gas Town v1.2.1 [5], [26]	Documented	Documented	Partial/configuration-dependent	Documented	Partial/configuration-dependent	Documented
Gas City v1.4.1 [6], [27]	Documented	Documented	Partial/configuration-dependent	Documented	Partial/configuration-dependent	Documented
DeepSeek Harness rc.2 [28], [29]	Documented	Documented	Partial/configuration-dependent	Documented	Documented	Not documented
Omnigent v0.10.0 [30], [33]	Documented	Documented	Documented	Documented	Documented	Partial/configuration-dependent

TABLE VI
AUTHORITY AND INTEGRITY PROPERTIES. NOT DOCUMENTED MEANS ONLY THAT THE REVIEWED PINNED PUBLIC CONTRACT DOES NOT STATE THE BINDING.

System	Writer identity	Incarnation fence	Exact verification subject	Effect idempotency/read-back	Protected integration	Truthful uncertainty
Bullet Farm design [4]	Documented	Documented	Documented	Documented	Documented	Documented
Bullet Farm Stage 1 [12], [19]	Partial/configuration-dependent	Partial/configuration-dependent	Partial/configuration-dependent	Partial/configuration-dependent	Partial/configuration-dependent	Documented
Gas Town v1.2.1 [5], [26]	Partial/configuration-dependent	Not documented	Not documented	Not documented	Partial/configuration-dependent	Partial/configuration-dependent
Gas City v1.4.1 [6], [27]	Partial/configuration-dependent	Not documented	Not documented	Partial/configuration-dependent	Partial/configuration-dependent	Partial/configuration-dependent
DeepSeek Harness rc.2 [28], [29]	Not documented	Not documented	Not documented	N/A	N/A	Partial/configuration-dependent
Omnigent v0.10.0 [30], [32]	Partial/configuration-dependent	Not documented	Not documented	Not documented	Partial/configuration-dependent	Partial/configuration-dependent

TABLE VII
PRIMARY ISSUE EVIDENCE MAPPED TO BULLET INVARIANTS AND CURRENT MATURITY.

Project / ID / state / version	Pinned-code or release corroboration	Failure class	Mapped Bullet invariant	Current maturity	Residual risk
Gas Town #4094; open; reported 1.1.0 [14]	v1.2.1 retains town/hook worktree architecture; issue is not treated as a pin-wide verdict	Repository/config aliasing	Sole writer; hostile Git/config refusal; private Attempt clone	Component-proved	Connected BulletGit authority and same-UID path races remain open.
Gas Town #742; closed; reported before v1.2.1 [35]	v1.2.1 was inspected; no “687 respawns” figure is present or claimed	Configuration placement	Sealed admitted HOME/config and exact executable subject	Component-proved	Live provider configuration and every real-provider receipt remain absent.
Gas City #1181; open; version not specified [15]	v1.4.1 documents rigs/packs; issue records parallel agents resolving to one rig path	Shared writable identity	Private per-Attempt clone; one writer principal	Component-proved	No connected positive BulletGit transaction.
Gas City #5511; open; affected v1.4.1 [16]	Exact selected release is the reported affected version; issue records post-commit timeout and duplicate workflows	Ambiguous effect retry	Durable effect identity; expected-old-OID CAS; read-back/adoption	Component-proved	No authenticated live forge receipt or transaction proof.
Gas City #1052; closed via #2985; affected 0.15.2 [17]	v1.4.1 is later than the reported fix; retained as historical class evidence only	Non-atomic claim	Atomic lease/fence/Attempt command transition	Component-proved	Signed online transport is not connected end to end.
Omnigent #1694; open; affected 0.2.0 [34]	v0.10.0/Polly materials document parallel worktrees and cross-vendor review	Session/config ambiguity	Durable Attempt state; incarnation fence; preservation before cleanup	Component-proved	Provider and reviewer identities are not yet live-bound in Bullet.
Omnigent #725; open; version not stated [36]	v0.10.0 scope was inspected; issue concerns external/native edits in a non-Git workspace	Incomplete change observation	Candidate manifest and honest UNKNOWN for unrepresented change state	Component-proved	Non-Git workspaces are outside Bullet V1; Git Candidate path is not connected.
Omnigent PR #4318; merged; included in v0.10.0 [8], [37]	Release notes identify the approval-attribution roll-back as a breaking change	Credential/approval attribution	Separate requester, approver, broker, and effect identities	Designed	Operation-specific components exist, but no live owner-credential effect proves separation.

- 2) crash before and after patch journal sync, Candidate publication, Evidence settlement, effect-intent sync, remote application, response receipt, read-back, preservation, deletion, and tombstone sync;
- 3) rebase or merge between verification and integration, change a gate executable or policy generation, and attempt to reuse stale Evidence;
- 4) inject hostile Git config, hooks, filters, alternates, attributes, credentials, helpers, submodules, symlinks, path normalization/case collisions, large frames, invalid Unicode, terminal escapes, and protocol contamination;
- 5) drop, duplicate, reorder, truncate, and delay every JSON-RPC and command response, including a server-applied forge write whose client response is lost;
- 6) replace an admitted pathname after validation, race cleanup targets, exhaust descriptor-relative traversal limits, and exercise each supported or explicitly refused

operating system; and

- 7) disconnect/reconnect the projection across sequence gaps, conflicting events, missing routes, and restore epochs, verifying that no display becomes authoritative.

The acceptance criterion is not merely “no crash.” Every injected boundary must end in one of a small set of independently inspectable outcomes: exact prior state, exact complete next state, durable failed state with preserved subject, or truthful UNKNOWN/quarantine requiring reconciliation. No outcome may contain two current fences, two logical effects, Evidence for the wrong Candidate, an unpreserved unique workspace, or a projection greener than its source.

Formal and implementation traces should cross-check each other without being confused. The two TLA+ models can generate small safety traces for lease/fence and effect ambiguity. Rust tests can replay those traces against component state machines. The connected corpus must then repeat equivalent

faults across real processes, filesystem boundaries, and a controlled forge. Agreement is stronger evidence than any layer alone, but it remains one offline Transaction-proved subject until live admission repeats the same binding.

C. Stage-2 Matched Benchmark

Stage 2 is prohibited until a signed connected TRANSACTION_PROOF exists. It will freeze identical repositories and issue tasks; pin every system, model, tool, policy, budget, and retry rule; run enough trials to expose tail failures; and retain exact final trees, tests, human adjudication, wall-clock time, token/cost measures, collisions, duplicated effects, ambiguous outcomes, recovery actions, and data loss. Operational success and authority violations will be separate dependent variables. Systems unable to emit a native receipt will be wrapped by an external observer without pretending that wrapper is a product feature.

Only that matched corpus can support “faster,” “cheaper,” “safer,” or “more successful.” This paper makes none of those claims. Public superiority language remains deferred with the benchmark.

Matched evaluation must also account for refusal. A system that declines every mutation has no integrity failures and no product utility; a system that always continues may look fast by discarding ambiguity. Results therefore report task utility, safe refusal rate, recovery latency, operator interventions, and unresolved unknowns together. Storage overhead from private clones, verifier reconstruction time, merge-queue dwell, and read-back traffic are first-class costs rather than hidden implementation details.

D. Falsification Conditions

The architecture would be weakened or rejected by any of the following:

- a stale or reclaimed incarnation mutates a repository or remote effect;
- two Attempts share a repository principal or common mutable control path;
- Evidence remains valid after any Candidate subject changes;
- timeout/restart can create two logical effects or erase an applied one;
- cleanup destroys the only recoverable bytes before preservation settles;
- Portal or prose can promote UNKNOWN to verified;
- the same guarantees are achieved by a substantially smaller public contract under the matched fault corpus; or
- bookkeeping/latency cost dominates benefit under realistic workloads.

The paper itself is falsified by a source row that does not support its cell, an unrecorded dirty-tree result, a snapshot mismatch, a non-reproducible PDF, or a claim above its maturity. Those are publication failures even if the Rust components remain correct.

E. Limitations and Artifact Statement

This is a design and component inventory from one date. Public documents can omit private controls; issues are selected counterexamples rather than incidence statistics; proprietary systems are black boxes; and open-source heads move after their pins. The V1 threat model does not solve multi-tenant remote execution, a malicious host kernel, semantic correctness of model output, non-Git workspaces, or every same-UID race. Private clones cost storage and setup time. Fencing, exact subjects, and read-back add state and latency. No connected-product measurement yet shows that the cost is worthwhile.

The artifact comprises this source, the executive brief, bibliography, architecture vector, evidence JSON and schema, generated macros, deterministic build/preflight scripts, and the non-authoritative workplan. The frozen V1 topology is Jeryu-only from immutable signed tags, with no authenticated URL or public mirror selected. The only permitted local Jeryu family is `/home/ubuntu/jain-split/jeryu-split`. The proposed `git.neverhuman.org` deployment and GitHub discovery mirrors remain operator-blocked and unratified. Jankurai remains its existing monorepo and requires preservation-first review before any mirror decision. None of those topology proposals is a live receipt [25].

VIII. CONCLUSION

Powerful agents increase the need for narrow, inspectable authority. Bullet Farm’s proposed contribution is neither a stronger persona nor another agent loop. It is the composition of familiar transaction mechanisms over exact Git subjects: fencing at mutation, one repository writer, private Attempts, immutable Candidates, independent Evidence, durable effect intent, read-back/adoption after ambiguity, protected integration, and truthful uncertainty across five planes.

The current evidence boundary is equally important. The family has meaningful component receipts and bounded models, while the connected transaction, every live provider/forge proof, and the release remain absent. The reviewed pinned public contracts do not document the same end-to-end bindings, but that is a contract comparison rather than measured superiority. Stage 1 publishes the architecture and its falsifiable gaps. Stage 2 begins only after the exact connected transaction exists.

APPENDIX A

DATED LANDSCAPE BEYOND THE DEEP STUDIES

Table VIII is an observation index, not a capability score. Open-source entries name the exact inspected commit; proprietary entries name official documents observed 2026-08-25 and remain black-box limited.

Release/commit pins are observation subjects, not endorsements and not claims about later versions. For proprietary systems, only the cited official surface is classified; undocumented properties remain Unknown.

TABLE VIII
LANDSCAPE SNAPSHOT OBSERVED 2026-08-25. FULL 40-HEX PINS ARE IN THE BIBLIOGRAPHY AND EVIDENCE MANIFEST.

System	Pinned subject or official observation	Public-contract scope and limitation
Codex	Official CLI/cloud docs [9], [10]	Local CLI and hosted delegated tasks are documented; public docs do not establish Bullet's connected authority chain. Proprietary/hosted behavior is black-box limited.
Claude Code	Official docs observed 2026-08-25 [38]	Agentic terminal/editor workflow; proprietary behavior and service controls are black-box limited.
Cursor	Official agent docs observed 2026-08-25 [39]	IDE/agent workflow; proprietary internals are black-box limited.
GitHub Copilot	Official coding-agent docs observed 2026-08-25 [40]	Hosted issue-to-PR workflow integrated with GitHub; service internals are black-box limited.
Jules	Official Google docs observed 2026-08-25 [41]	Hosted asynchronous coding tasks; service internals are black-box limited.
Devin	Official docs observed 2026-08-25 [42]	Hosted coding-agent sessions; service internals are black-box limited.
Kiro	Official docs observed 2026-08-25 [43]	IDE/spec-driven agent workflow; service internals are black-box limited.
Gemini CLI	commit 812f7a2bcf20 [44]	Open-source terminal agent; Apache-2.0.
OpenHands / Agent Canvas	commit 150e76046db0 [45], [46]	Open agent platform and control surface; MIT. Agent Canvas documents isolated worktree operation and ACP composition.
SWE-agent	commit 3ea751c087f3 [47]	Research issue-solving harness; MIT.
mini-SWE-agent	commit 25941c89cfeb [48]	Minimal research harness; MIT.
Aider	commit 5dc9490bb35f [49]	Open terminal pair-programmer; Apache-2.0.
Cline	commit 4f5f2384076e [50]	Open editor agent; Apache-2.0.
Continue	commit 5522c6f44ca0 [51]	Open IDE/agent framework; Apache-2.0.
Goose	commit f1d32e8dbbdf [52]	Open extensible agent; Apache-2.0.
Roo Code	commit b867ec914575 [53]	Open editor agent; Apache-2.0.
ACP	commit 5acaa3acdall [54]	Agent-client interoperability protocol; Apache-2.0; not a repository authority system.
MCP	commit 57ac4a2ec742 [55]	Model-tool/context interoperability protocol; MIT; not a repository authority system.

APPENDIX B

EVIDENCE MANIFEST AND PUBLICATION PREFLIGHT

The machine source is `docs/paper/evidence.json`; its schema identifier is `bullet.paper-evidence.v1`. Eight required groups record the snapshot ID, UTC time and source epoch; exactly four repository commit/tree/clean subjects; tool versions; full commands with exit code, semantic outcome, and limitation; shared result values; the fixed four-term maturity vocabulary; dated external pins; and hashes of evidence-critical inputs. The build validates that object and emits `evidence.generated.tex`, which both documents consume. Gate counts, repository abbreviations, audit results, and absent transaction status are therefore not manually duplicated.

PDFs do not appear in their own input manifest: embedding a PDF hash in source that changes the PDF creates a circular fixed-point problem. Successful preflight reports final hashes as output receipts. The Hub commit has a related self-reference boundary. The snapshot binds the clean evidence subject before publication; a later Hub head is accepted only when it descends from that subject, every intervening path belongs to the enumerated paper/tooling set, and hashed evidence inputs are unchanged. Kernel, BulletGit, and Portal must match exactly. This narrow exception cannot lend old receipts to code changes.

Strict preflight rejects a dirty capture or checkout, false maturity, repository/commit/tree mismatch, unexpected Hub change, stale input hash, undefined/multiple citation or reference, duplicate or unused bibliography entry, and an external reference without URL/DOI and access date. It builds twice under one `SOURCE_DATE_EPOCH`, UTC timezone, deterministic PDF metadata, and locale; both PDF hashes must repeat. Layout/PDF checks require no material overfull box, US Letter, title/author/subject/keywords, embedded fonts, 12–14 paper pages, and exactly four brief pages. Every rendered page is then inspected because no log parser proves that a table, URL, vector arrow, or final column is visually legible.

This is publication assurance, not a product gate. A byte-identical paper can faithfully report a broken product, and a correct component can coexist with a failed paper check. The guarantee is narrower: both documents state the same machine-backed facts about the same subjects and common accidental overclaim fails closed.

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